

CampusFix AI

Visual Campus Maintenance & Resolution Intelligence System

"Don't fill a complaint form. Show the problem."

PROJECT DOMAIN Campus Physical Infrastructure	HARDWARE REQUIREMENT Zero Capex (Standard Smartphone)	AI CORE ENGINE Multimodal Vision + LLM Orchestration	PRIMARY INNOVATION AI-Powered Before/After Verification
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1 Problem Statement: The Campus Maintenance Gridlock

Educational institutions manage vast physical estates comprising hundreds of lecture halls, laboratories, hostels, washrooms, and common corridors. Across these facilities, physical maintenance failures occur continuously: flickering lights, broken ceiling fans, shattered window panes, leaking plumbing, loose electrical sockets, and damaged furniture. While institutions maintain dedicated facilities and maintenance departments, the administrative interface between students and service teams is fundamentally broken.

Current systems fail due to four structural bottlenecks:

- High Reporting Friction & Vague Descriptions** Students must navigate tedious fields, select hierarchical categories, or guess asset codes. In response, students submit vague verbal or text complaints like 'fan not working in Block B', forcing technicians to inspect blind without knowing what tools or spare parts are needed.
- Departmental Misrouting** Students do not know whether an issue belongs to Electrical, General Maintenance, IT, or Civil Engineering. A malfunctioning interactive projector bounces between IT and Electrical for days before anyone takes ownership.
- Duplicate Ticket Floods** For common issues (such as a water cooler or washroom pipe), dozens of students independently file complaints. This multiplies ticket backlogs without multiplying usable information, paralyzing administrative staff.
- The Phantom Resolution Problem** The current system is lack of accountability. Tickets are marked 'Resolved' via a simple checkbox on an admin screen, with zero physical proof that the issue was actually addressed. Students are left frustrated when problems persist.

2 The Core Idea: An Evidence-Based Digital Lifecycle

CampusFix AI replaces manual forms with an AI-first visual pipeline. The core premise is simple: 'Don't fill a complaint form. Show the problem.' A student or faculty member snaps a single photo. From this visual evidence, the system autonomously identifies the object, detects the defect, assesses urgency, extracts location context, and routes a structured work order to the exact technician required. Critically, resolution is verified through visual diffing between before and after repair photographs.

1. CAPTURE -> 2. UNDERSTAND -> 3. AUTO-ROUTE -> 4. REPAIR -> 5. VERIFY -> 6. CLOSE

3 The Solution Approach & Architecture

CampusFix AI is architected as a Dual-Engine Intelligent System that bridges computer vision perception with enterprise workflow automation. It operates without proprietary IoT hardware, relying on modern multimodal foundation models.

ENGINE A: Visual Perception & Intelligence	ENGINE B: Workflow Automation & Verification
- Zero-Shot Defect Detection: Identifies broken fans, leaks, cracked glass, damaged furniture. - Room Scan Mode: Multi-object spatial parsing from a single panoramic photo. - OCR Evidence Extractor: Extracts room numbers, block signage, and asset badges. - Hazard Triage: Evaluates visible safety risks (exposed wiring, flood and)	- Intelligent Routing: Automatic classification to Electrical, Plumbing, IT, Carpentry. - Duplicate Fusion: Similarity engine merges redundant reports for the same asset. - Before/After Diff Engine: Multimodal verification comparing repair against incident. - Resolution Evidence Store: Audit trail with timestamps, staff ID, and photos.

4 How It Works: Step-by-Step Incident Lifecycle

Step 1: Visual Ingestion	The user opens the mobile web app and captures a photo of the defect. Users can either snap a focused close-up or use Room Scan Mode from the entrance.
Step 2: Multimodal Scene Analysis	The Gemini Vision engine decomposes the image into affected objects, visible defects, failure severity, and ambient room context. Text in the background is parsed via OCR for room number and block name.
Step 3: Duplicate Detection Check	Before creating a new record, the system cross-references open tickets using location proximity, issue category, semantic text similarity, and visual embedding distance. If a match is detected, the user is offered a 1-tap 'Upvote & Track' option.
Step 4: Automated Departmental Routing	The confirmed defect is routed to the designated facility department (e.g. Broken Fan -> Electrical; Leaking Faucet -> Plumbing). A structured maintenance task is dispatched with full visual context and priority score.
Step 5: Physical Repair & After-Photo Upload	The assigned maintenance technician executes the physical repair. To close the work order, the technician is required to photograph the repaired asset from the same vantage point.
Step 6: AI-Assisted Verification & Closure	The Vision AI performs a comparative evaluation between the 'Before' and 'After' photos, scoring resolution consistency (e.g. 'Fan blades intact, motor reassembled - 96% Consistent'). The supervisor conducts final sign-off.

5 What Makes CampusFix AI Special: Hero Innovations

HERO INNOVATION #1: The Before/After AI Verification Loop (Proof of Resolution)

In physical maintenance, the root cause of distrust is the "phantom closure" -- tickets marked completed when work was either half-done or ignored. CampusFix AI enforces a mandatory proof-of-work protocol: a work order cannot transition to 'Resolved' without an 'After' photograph. The Multimodal Vision AI performs a comparative spatial and semantic diff between the original evidence and the post-repair evidence. It verifies that the defective component has been replaced or repaired, confirms that environmental features (wall paint, background fixtures) match the original location, and produces a structured confidence assessment. This gives administrators 100% auditable proof of physical resolution.

HERO INNOVATION #2: Room Scan Mode (Multi-Defect Spatial Intelligence)

When an entire classroom or lab has multiple problems (e.g., three broken lights, a fractured bench, and an unmounted switchboard), users will not file three separate forms. Room Scan Mode allows the student, faculty member, or facility inspector to take a single wide-angle photo from the doorway. The AI identifies and tags every candidate defect in one pass, categorizes each by department, and presents a multi-item checklist. The user can review, uncheck false positives, and submit all confirmed incidents simultaneously with one tap.

6 Comprehensive Feature Matrix

Capability / Feature	Operational Functionality	Direct Benefit to Campus
Visual Issue Detection	Zero-shot visual defect recognition using multimodal LLM	Eliminates forms; 80% reduction in reporting friction
Room Scan Multi-Audit	Identifies multiple defects from a single wide classroom photograph	Full room inspection completed in under 15 seconds
Duplicate Fusion Engine	Merges redundant reports using location, category & visual similarity	Prevents queue flooding; eliminates duplicate dispatch
Auto-Routing Dispatcher	Maps defects to Electrical, Plumbing, IT, Carpentry, Housekeeping	Eliminates misrouted tickets bouncing between departments
Before/After Verification	Comparative AI image analysis verifying repair completion	Zero phantom closures; creates verifiable proof-of-work
Safety & Hazard Scoring	Detects emergency risks (exposed wires, water near power)	Automatically elevates high-risk issues to Priority 1
OCR Metadata Extraction	Reads room numbers, wing banners, and equipment labels from photos	Automatically binds tickets to specific rooms/assets
Unified Admin Dashboard	Live overview of open, assigned, in-progress, and verified repairs	Transparent workload tracking & facility audit readiness

7 Technical Architecture & System Specifications

System Layer	Technology Selection	Design Rationale
Frontend UI	React + Vite + Tailwind CSS + Lucide	Ultra-fast progressive mobile web app; zero app store friction
Backend REST API	FastAPI (Python) / Node.js	Asynchronous request handling, rapid AI multimodal orchestration
Multimodal AI Core	Google Gemini Vision API	Zero-shot visual reasoning; eliminates custom CV training cycles
Data & Evidence Store	SQLite / PostgreSQL + Cloud Storage	Structured relational incident data + secure image repository
Security & Access	Role-Based Access Control (RBAC)	Segregated portals for Students, Technicians, and Administrators
Enterprise Integration	RESTful Webhooks & JSON APIs	Seamless future integration with ERPs (SAP, Peoplesoft, custom)

8 Projected Campus Impact & Accreditation Value

Operational Dimension	Traditional Campus Process	CampusFix AI Target	Institutional Impact
Incident Logging Time	3 to 5 minutes (forms & calls)	Under 10 seconds (photo snap)	Higher student engagement & reporting compliance
Redundant Tickets	30% to 50% duplicate reports	Filtered & merged dynamically	Administrative workload cut by nearly half
Department Triage	Manual dispatch (12-48 hours)	Instant auto-routing	Faster turnaround time from defect to dispatch
Proof of Resolution	Anecdotal / checkbox only	Auditable before/after imagery	Eliminates unresolved recurring complaints
Institutional Audit	Scattered paper / manual logs	Structured digital incident logs	Direct evidence for NAAC / ABET accreditation

9 Executive Summary of the Project Vision

CampusFix AI bridges the longstanding divide between the physical reality of campus maintenance and administrative workflows. By combining zero-shot multimodal vision, intelligent deduplication, and rigorous before/after verification, it eliminates paperwork, ends duplicate tickets, and guarantees that reported problems are verifiably fixed. It requires zero hardware expenditure and delivers immediate operational visibility for students, staff, and campus leadership.

"CampusFix AI turns physical campus maintenance into an evidence-based, transparent digital lifecycle."